
Global CO₂ Emissions: Who Emits, How Much, and What's Changing

Fossil fuels & industry, 1750–2024

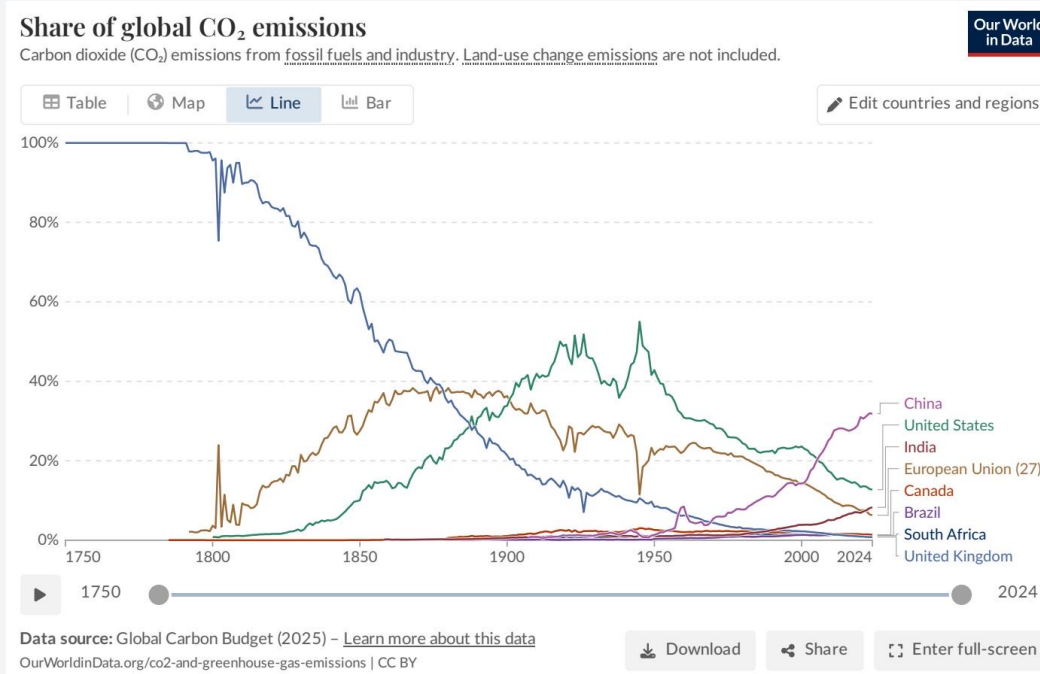
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CONFIDENTIAL

KEY INSIGHT Emissions have surged since 1950 and have not yet peaked.

- Slow pre-industrial growth followed by a dramatic post-1950 acceleration
- Recent slowing in growth rate, but global total has not yet peaked



02 Europe and the US Fell from 90% of Emissions to Under One-Third

Global CO₂ Emissions

KEY INSIGHT The geographic center of emissions has fundamentally shifted to Asia.

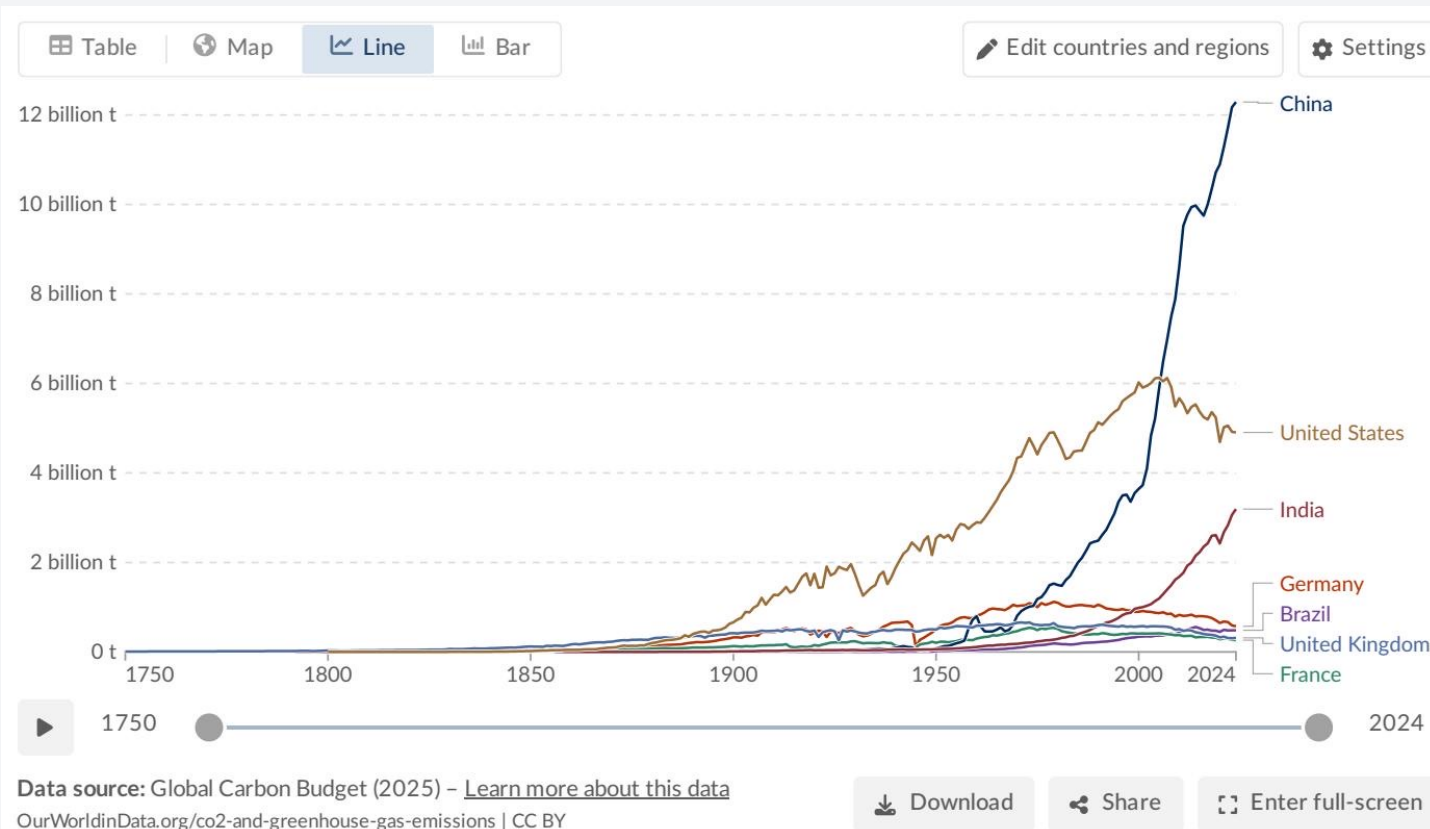
- In 1900, Europe and the US were responsible for >90% of global emissions
- Today, their combined share is under one-third
- Asia (led by China) has emerged as the dominant source

[Stacked Area Chart Placeholder: Regional Share Shift]

KEY INSIGHT China emits more than the US, India, and the EU combined.

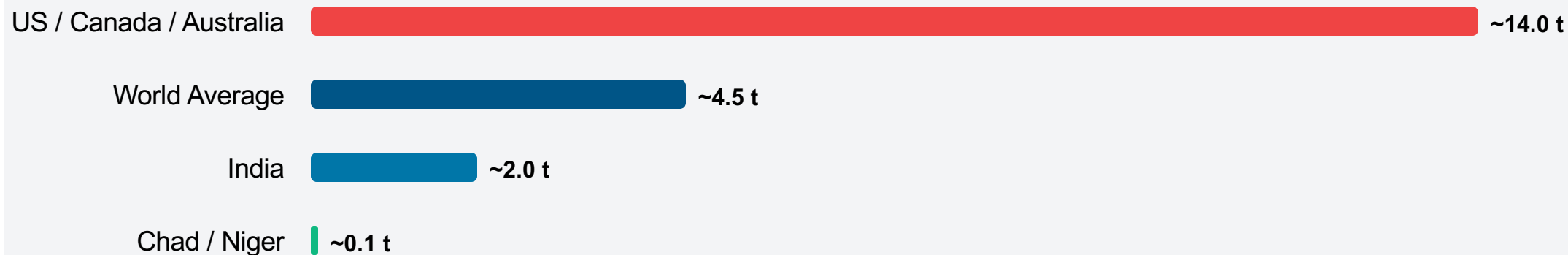
2024 Share of Global CO₂:

- 1. China: ~31%**
Rising, but growth slowing
- 2. United States: ~13%**
Actively declining
- 3. India: ~8%**
Rising steadily
- 4. EU-27: ~6%**
Actively declining



KEY INSIGHT Absolute emissions mask profound per capita inequalities across nations.

- A roughly 150x gap exists between the lowest and highest emitting nations
- The average American or Australian emits in under two days what the average person in Sub-Saharan Africa emits in an entire year

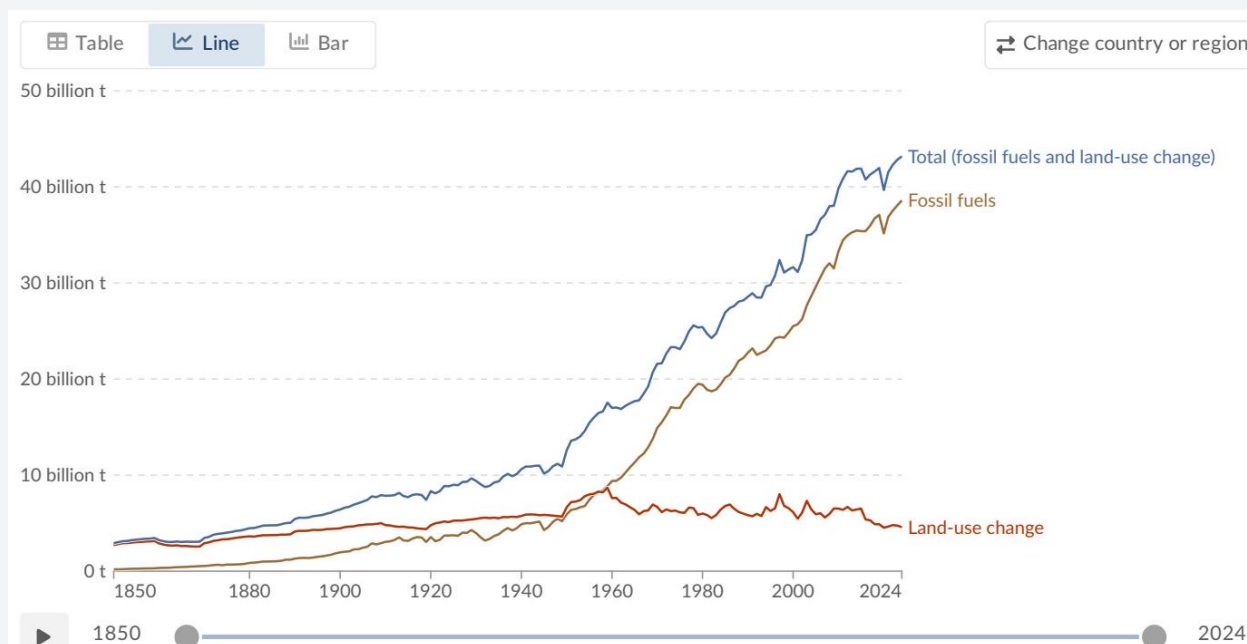


05 Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China

Global CO₂ Emissions

KEY INSIGHT Balancing emission reductions with the right to economic development.

- While China is the largest absolute emitter, its per capita emissions (~9 t) are still well below the US and Canada (~14 t).
- India remains very low on a per capita basis at roughly 2 t.

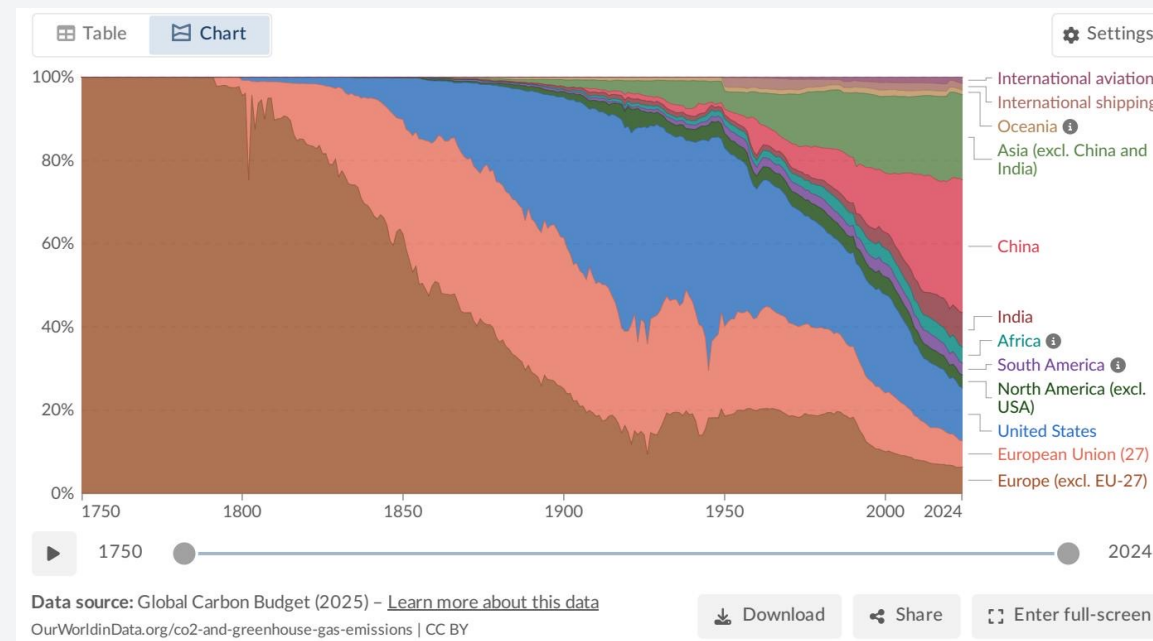


Source: Global Carbon Budget 2025 via Our World in Data

Analysis for policy and climate negotiations

KEY INSIGHT The global transition is happening, but it is highly uneven geographically.

- Many Western nations (US, UK, much of EU) saw their emissions decline in 2024.
- These gains were offset by increases in Russia, SE Asia, and parts of Africa.
- Central Asia experienced increases exceeding 10%.

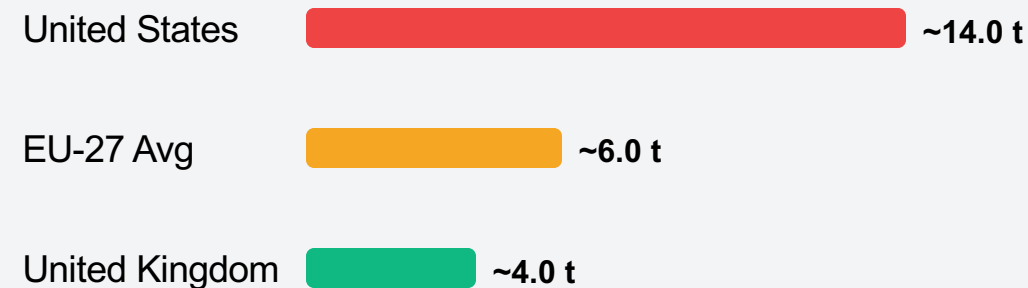


 **KEY INSIGHT** Deliberate energy policy and technology choices drive emissions down.

Why the difference?

- High prosperity does not mandate high emissions.
- France, UK, and Portugal have successfully decoupled economic output from fossil fuel consumption.
- Driver: A much higher share of electricity comes from nuclear and renewable sources.

Per Capita Emissions (Wealthy Nations)



KEY INSIGHT

The path forward requires addressing both total volume and deep per capita inequities.

1. Global Volume is Unprecedented

Global emissions exceed 35 Bt/yr and haven't peaked.

2. Asia Has Become the Core Source

China is the largest annual emitter (~31%), but its per capita footprint is still below the US.

3. Extreme Inequalities Persist

Per capita gaps of 150x remain between the richest and poorest nations.

Historical responsibility remains dominated by Europe and the US.

4. Energy Policy Matters

Clean energy technology choices can successfully decouple prosperity from carbon emissions.